

Reman Dey

Engineering Physics, IIT Mandi

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Summary

Engineering Physics student at IIT Mandi. Passionate about research-oriented physics problems and hands-on engineering, with experience spanning laser optics, robotics, embedded systems, and AI/software development, from building functioning hovercrafts and laser transceivers to developing AI-powered chatbots, ML models, and autonomous rover navigation systems.

Education

Indian Institute of Technology (IIT) Mandi

India

B.Tech in Engineering Physics

Expected Graduation: 2029

CGPA: 9.74

Technical Skills

- **Languages & Frameworks:** Python, C++, C, ROS2 Humble, Nav2, PX4, JavaScript, HTML5, CSS3, React
- **Hardware & Embedded:** STM, ESP8266, ESP32, Arduino, Raspberry Pi, Robotics, Electronics, PCB Prototyping
- **AI & Software:** Image Processing, Machine Learning, NLP, RAG, Google Gemini API
- **Tools:** Git, GitHub, SSH, Linux

Experience

Weather Models For Himalayan Region

IIT Mandi

Internship under Prof Sarita Azad

May 2026 – July 2026

- Trained a Weather Model based on the PINN Architecture to predict Temperature

Centre for Human-Computer Interaction (CHCi)

IIT Mandi

Internship

June 2026 – July 2026

- Analyzed EMG Data to classify hand gestures for controlling a mobile rover

Team Deimos — IIT Mandi Mars Rover Team

IIT Mandi

Software Team Member

May 2026 – Present

Link: <https://github.com/Team-Deimos-IIT-Mandi/rovercore>

- Wrote a complete fuel trail following script from scratch, using polynomial fitting of the fuel trail to enable predictive path planning for the rover.
- Developed the entire codebase for autonomous hill peak detection within a geofenced region, with 0.5 meters positional accuracy.
- Trained a machine learning model for basalt rock detection, achieving 76% classification accuracy, to support the rover's geological survey capabilities.
- Designed and implemented a custom path planner based on Artificial Potential Fields (APF) after Nav2 proved infeasible on the rover's limited onboard compute resources, reducing compute overhead.

Achievements

- Secured **2nd position** in the Zentej Hackathon (AI/ML problem statement) for building a model to detect AI-generated faces from real ones.
- Ranked **14th** in IIT Bombay Techfest's theoretical submission on Aircraft Details, among 200 selected candidates.
- Achieved a **Dice score of 0.69** in the Solar Flare Filament Segmentation Kaggle competition.

Projects

Open-Source Micro-Rover

Embedded Systems / *May 2026* / *GitHub*

- Designed and open-sourced a low-cost micro-rover platform with embedded control and mechanical design, enabling students and hobbyists to build and extend their own rovers.

JoSAA Counselling RAG Chatbot

Python, RAG, LLMs / *May 2026* / *GitHub* | *Live Demo*

- Built and deployed a Retrieval-Augmented Generation chatbot that answers student queries on IIT Mandi's JoSAA counselling process, grounding responses in official counselling documentation to ensure factual accuracy.

PDF Question Generator

Python, LLM / *June 2026* / *GitHub*

- Built an LLM-powered tool that automatically extracts key content from PDF documents and synthesizes practice questions, streamlining exam preparation and content review.

Hovercraft

Physics / Mechanical Engineering / *2018*

- Designed and built a functioning hovercraft from first principles of fluid dynamics.

Laser Transceiver

Electronics / Optics / *2016*

- Built a laser-based transmitter-receiver pair for free-space optical communication.

Tesla Turbine

Physics / Mechanical Engineering / *2020*

- Constructed a bladeless turbine exploiting viscosity and boundary layer effects.

DodoTalks

Web, Google Gemini API / *May 2025* / *Live*

- Developed an AI-powered chat application using the Google Gemini API with a custom conversational interface.

Areas of Interest

- Advanced Physics Research & Experimental Design (laser labs, instrumentation)
- Robotics & Intelligent Automation (embedded + cognitive systems)
- Software & AI Development (NLP, bots, real-world AI applications)
- Hardware Prototyping & Embedded Systems (PCB to physics-machines)